RAG Assignment Report: Baseline Retrieval vs. Cross-Encoder Re-Ranking

GitHub Repository Link: https://github.com/Moiesha/RAG

Dataset Used: 2025 FIA Formula 1 Sporting Regulations (Issue 5, 119 pages, split into 779 overlapping chunks of 115 words with a 30-word overlap).

Stage 1 Bi-Encoder: sentence-transformers/all-MiniLM-L6-v2 (384-dimensional normalized embeddings indexed in FAISS IndexFlatIP).

Stage 2 Cross-Encoder: cross-encoder/ms-marco-MiniLM-L-6-v2 (re-ranking Top 10 candidates into Top 3).

Generator LLM: Qwen/Qwen2.5-0.5B-Instruct (local open-weights instruction model).

1. Architecture Comparison: Bi-Encoder vs. Cross-Encoder

Bi-Encoder (Independent Embeddings & Cosine Similarity): A Bi-Encoder processes the user query and each document chunk separately. Every chunk in the PDF is passed through the Transformer offline and compressed into a single fixed-size 384-dimensional vector. At query time, the query is encoded into its own 384-dimensional vector, and similarity is computed using cosine similarity (inner product of unit-normalized vectors). Because each document chunk is compressed into a static vector without knowing what the query will ask, fine-grained details such as negations ("not permitted"), conditional exceptions ("during the last three laps"), and cross-chunk table references get averaged out into a general topical vector.

Cross-Encoder (Joint Self-Attention Across Tokens): A Cross-Encoder does not produce separate embeddings. Instead, it concatenates the query and a candidate chunk into a single token sequence ([CLS] query [SEP] chunk [SEP]) and passes them together through the Transformer layers. This enables full joint self-attention across every query token and every document token simultaneously at every layer. As a result, the Cross-Encoder directly attends between specific qualifiers in the query and exact regulatory clauses in the passage, rather than relying on a single compressed summary vector.

2. Qualitative & Rank Shift Analysis

To evaluate where Baseline Retrieval fails and Cross-Encoder re-ranking restores relevance, four regulatory stress-test queries were evaluated over the 779-chunk corpus. In every case below, the critical regulatory chunk was buried between Rank #4 and Rank #10 in Stage 1, and promoted into the Top 3 by the Cross-Encoder:

Query & Failure Category	Vital Chunk(s)	Baseline Rank (Bi Score)	Re-Ranked Pos (CE Score)	Rank Shift
Q1: Exceptions & Corner Cases (5s/10s penalty in final 3 laps)	Chunk #435 (p. 63) Chunk #434 (p. 63)	Rank #9 (0.6157) Rank #10 (0.5994)	Rank #2 (4.2786) Rank #3 (3.7658)	+7 (Promoted) +7 (Promoted)
Q2: Negation & Constraints (Mechanics working during penalty)	Chunk #438 (p. 64) Chunk #440 (p. 64)	Rank #8 (0.6088) Rank #5 (0.6457)	Rank #1 (4.7390) Rank #2 (3.9907)	+7 (Promoted) +3 (Promoted)
Q3: Keyword Trap / Distractor (Pre-Race vs. Post-	Chunk #344 (p. 52) Distractor: #511 (p. 73)	Rank #4 (0.5763) Distractor at #1 (0.6271)	Rank #1 (5.6698) Dropped to #5 (0.0240)	+3 (Promoted) -4 (Demoted)

Race Parc Fermé)				
Q4: Document-Specific Failure (Shortened race <25% points table)	Chunk #33 (p. 6)	Rank #9 (0.5833)	Rank #3 (0.6441)	+6 (Promoted)

Comparison of LLM Answer 1 (Baseline Top 3) vs. Answer 2 (Re-Ranked Top 3)

- **Q1 — Exceptions & Corner Cases (5s/10s Penalty in Final 3 Laps):** The query asks why the pit stop serving rule does not apply when a 5-second or 10-second time penalty is imposed during the final three laps of a race, and what happens instead. Baseline Top 3 retrieved Chunks #431, #432, and #433, which only state the general rule that a driver must enter the pit lane and stop at their next pit stop. The specific exception for the last three laps (Chunk #435, Article 54.4c: stating that Article 54.4b does not apply in the last 3 laps and 5s/10s are added to elapsed race time instead) sat at Rank #9, while Chunk #434 sat at Rank #10. Because Chunk #435 was absent from the Top 3, Answer 1 hallucinated an incorrect rule from Chunk #433 (claiming the exception is for an emergency stop or safety car procedure). After the Cross-Encoder promoted Chunk #435 (Rank #9 → Rank #2) and Chunk #434 (Rank #10 → Rank #3), Answer 2 correctly stated that in the final three laps the driver does not need to make another pit stop to serve the penalty.
- **Q2 — Negation & Constraints (Mechanics Working on Stationary Car):** The query asks what team mechanics are strictly not permitted to do while a car is stationary in the pit stop position serving a time penalty. The Bi-Encoder ranked Chunks #432, #431, and #433 at Ranks #1–#3 because they share high vocabulary overlap around pit stop time penalties, but none of them state the prohibition on mechanics. The Cross-Encoder attended directly to the negation constraint and promoted Chunk #438 (Rank #8 → Rank #1, score 4.7390) and Chunk #440 (Rank #5 → Rank #2, score 3.9907), which state that while stationary serving a time penalty, the car may not be worked on (except restarting a stalled engine after the penalty period has elapsed). Answer 1 merely repeated the question back (saying mechanics cannot stop the car), whereas Answer 2 captured the stalled-engine work restriction from Chunk #440.
- **Q3 — Keyword Trap / Distractor (Pre-Race vs. Post-Race Parc Fermé):** The query asks why a competitor is not allowed to modify aerodynamic or suspension settings after qualifying under Parc Fermé regulations unless climatic conditions change. Because Chunk #511 (Article 60, Post-Race Parc Fermé on page 73) contains the exact phrase "parc fermé" and "No intervention of any kind is allowed there", the Bi-Encoder fell for the keyword overlap and placed Chunk #511 at Rank #1 (0.6271), pushing the actual Pre-Race Parc Fermé setup rule (Chunk #344, page 52, Article 40.9) down to Rank #4 (0.5763). The Cross-Encoder recognized that Chunk #511 did not address aerodynamic/suspension setup or climatic changes, demoting Chunk #511 from Rank #1 → Rank #5 (score 0.0240) and promoting Chunk #344 from Rank #4 → Rank #1 (score 5.6698).
- **Q4 — Other Document-Specific Failure (Shortened Race Distance Points Table):** The query asks how many championship points are awarded if a race is suspended after more than two laps but less than 25% of the scheduled distance. Chunk #31 (Rank #1) contains the text rule referring to Column 1 of the table on the next page, but the actual reduced-distance Race points table (Chunk #33, page 6: awarding 6, 4, 3, 2, 1 points for 1st–5th) consists mostly of numbers and received a low Bi-Encoder similarity score (Rank #9, 0.5833), while the Sprint points table (Chunk #30: 8, 7, 6...) sat at Rank #2. The Cross-Encoder promoted Chunk #33 from Rank #9 → Rank #3 (+6 positions), placing the true reduced points table into the Top 3 context. (An interesting observation from our run is that because Chunk #31 starts with the

clause "no points will be awarded unless a minimum of two laps have been completed", the small 0.5B LLM latched onto "no points" in both answers—showing that even when Cross-Encoder re-ranking successfully rescues a numerical table from Rank #9 to Rank #3, a small generator model can still struggle to join prose and tables across chunk boundaries.)

3. Latency Profiling

The execution time of Stage 1 Baseline Retrieval (Bi-Encoder query encoding + FAISS inner-product search over 779 chunks) vs. Stage 2 Cross-Encoder Re-Ranking (scoring all 10 candidate pairs on CPU) was recorded across all four queries:

Query ID	Failure Category	Baseline Retrieval (ms)	Cross-Encoder Re-Rank (ms)	Total Re-Ranking Pipeline (ms)
Q1	Exceptions & Corner Cases	21.36 ms	583.79 ms	605.14 ms
Q2	Negation & Constraints	50.51 ms	255.26 ms	305.78 ms
Q3	Keyword Trap / Distractor	17.99 ms	353.20 ms	371.19 ms
Q4	Other Document-Specific Failure	29.54 ms	335.30 ms	364.84 ms
MEAN	**Average Across Queries**	**29.85 ms**	**381.89 ms**	**411.74 ms**

Runtime Observation: On CPU, Baseline Retrieval averaged 29.85 ms per query because all 779 chunk embeddings were pre-computed offline. Adding Cross-Encoder re-ranking over the 10 candidates took an average of 381.89 ms (bringing total retrieval latency to 411.74 ms). Adding ~0.38 seconds per query is a minor latency overhead compared to LLM generation time, while rescuing critical rules that were buried at Ranks #4–#10.

4. Scaling Question: Why Not Run a Cross-Encoder Across the Entire Collection From the Start?

- **1. Document Representations Cannot Be Pre-Computed Offline:** In a Bi-Encoder, all N document chunks are embedded once ahead of time and stored in a vector index (FAISS). At query time, the model only encodes the single query string and performs fast vector dot products. A Cross-Encoder, however, requires the query tokens and document tokens to be concatenated together ([CLS] query [SEP] chunk [SEP]) before entering the Transformer so that self-attention can operate across both simultaneously. Because the query is unknown until runtime, nothing can be pre-indexed.
- **2. Linear O(N) Transformer Forward Passes at Query Time:** Running a Cross-Encoder across an entire collection of N chunks requires N full Transformer forward passes for every single user query. In our notebook benchmark on this single PDF corpus (N = 779 chunks), Two-Stage Re-Ranking (Top 10 chunks) took 411.74 ms (0.41 seconds), whereas running the Cross-Encoder across all 779 chunks took 22,524.27 ms (22.52 seconds)—making full-corpus Cross-Encoder scoring 54.7× slower even on a single document. If the collection scaled to a realistic corpus of 100,000 chunks, running a Cross-Encoder from the start would take roughly (100,000 / 779) × 22.52 s ≈ 2,890 seconds (around 48 minutes per query). The two-stage architecture solves this by using the fast Bi-Encoder to filter N chunks down to the Top 10 in ~30 ms, so the Cross-Encoder only runs its expensive joint self-attention on 10 pairs.